

Pupillary oscillation induced by pseudo-isochromatic stimuli for objective color vision test

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ABSTRACT

Color vision tests are conducted based on subjective responses; however, since subjective responses can be biased and are not uniformly applicable to all individuals, an objective method is required. In this study, we developed a quantitative and user-friendly color vision test, which utilized a psychophysical method based on pupillary responses to color flicker stimuli sampled from the confusion line. The results revealed that power spectral density of pupillary oscillations increased corresponded to the color differences. Our system could eventually lead to a feasible color testing method.

CCS CONCEPTS

• **Applied computing** → *Life and medical sciences*; • **Human-centered computing** → *HCI design and evaluation methods*.

KEYWORDS

color perception, pupillary response, color vision test, color vision deficiency

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1 INTRODUCTION

In previous studies, visual sensitivity ratios for each color vision characteristic were determined by the flicking of different color combinations and the adjustment of the light radiance ratio by the participants (e.g., [Ikeda and Urakubo 1968]). However, conventional color vision tests are based on observers' subjective responses. Therefore, observers with motor disabilities may have difficulties with behavioral adjustment tasks. Furthermore, a child's age, coordination, and developmental conditions hinder obtaining subjective

ratings in pediatric testing [Tekavčič Pompe 2020]. To address these issues, we focused on the pupillary response; a physiological and objective metric. The pupillary response changes by the light and its magnitude reportedly varies by the wavelength of the light [Mathôt 2018].

This study aimed to establish a quantitative and simple assessment method for color vision testing based on pupillometry, and to investigate the relationship between pupillary response and human color perception.

2 METHODS

2.1 Participants

Eleven individuals participated in the experiment (11 men; age range: 22–30 [$M = 24.0$, $SD = 2.34$]). All experimental procedures and methods followed the ethical guidelines stipulated in the Declaration of Helsinki and were approved by the Ethics Committee for Human Research at the Toyohashi University of Technology. Participants provided written informed consent, and the experiment was conducted in accordance with the guidelines of the committee.

2.2 Stimuli and Apparatus

Equiluminant pseudo-isochromatic color patch were used as stimuli. These color patch were inspired by the "Ishihara Plates," which are commonly used for color vision tests. These stimuli were flickering with a two-pair of colors sampled from red/green pseudo-isochromatic stimuli: colors that are indistinguishable among people with deutan color vision. The averaged luminance of the stimuli was 61.8 cd/m^2 , and the luminance differences between all pairs of stimuli were less than 1.0 cd/m^2 . The stimuli totalled 11, including ten flickering red/green pairs and one non-flickering stimulus

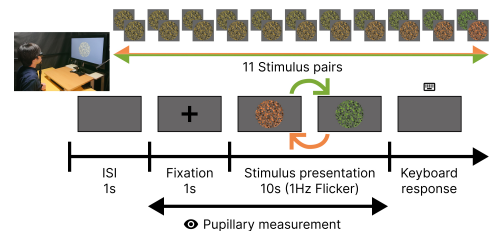


Figure 1: Experiment procedure and stimulus pairs.

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(Fig. 1). Experimental stimuli were presented on Display++ (Cambridge Research Systems Ltd., Rochester, UK) with $1,920 \times 1,080$ pixels and refresh rate at 120 Hz. Pupillary responses were measured binocularly using EyeLink 1000 Plus (SR Research Ltd., Oakland, Canada) at a sampling rate of 500 Hz. MATLAB R2021a (The MathWorks, Inc., Natick, Massachusetts, USA) and Psychtoolbox 3 were used for stimuli presentation.

2.3 Procedure

A fixation cross was presented on the monitor, followed by a pseudo-isochromatic stimulus flickering at 1 Hz for 10 seconds. A total of 55 trials of 11 stimulus pairs were conducted, five for each condition. The experimental condition were randomly selected for each trial, and the participants answered two questions after flicker presentation: First, participants answered whether they perceived a change in the color, and then, the perceptual intensity of the flickering using a visual analog scale (range: 0–100). The magnitude of the color difference was defined as the flickering intensity, and the task of this experiment was conducted after the first task in which the participants were asked to remember the highest and lowest flicker intensities. The participants were instructed to refrain from blinking during the fixation and stimulus presentation. All trials were divided into three sessions with breaks inbetween, and a 5-point calibration of the eye tracker was performed before the experimental task. Pupillary measurement data were interpolated by preprocessing, and trials exceeding the rejection rate were excluded from further analyses.

3 RESULTS

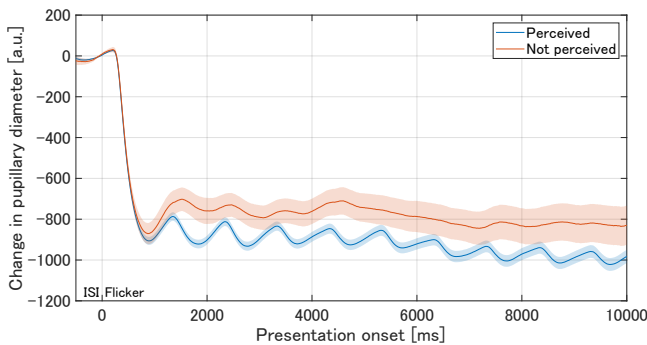


Figure 2: Results of pupillary response. Change in grand averaged pupillary diameter during stimulus presentation. Error bars represent standard error of the mean.

Fig. 2 shows the pupillary responses to each perceptual conditions (perceived, not perceived). The pupillary response to the flicker stimuli was oscillatory with a larger amplitude when the flicker was visible. Therefore, we computed power spectral density (PSD) from pupillary responses and performed a one-way ANOVA on the stimulus conditions. The result revealed a significant difference between the stimulus conditions ($F[10, 100] = 8.811$, $p < 0.001$, $\eta_p^2 = 0.468$); a post hoc t -test was also conducted. The post hoc analysis revealed * significant differences between several conditions (Fig. 3(a)).

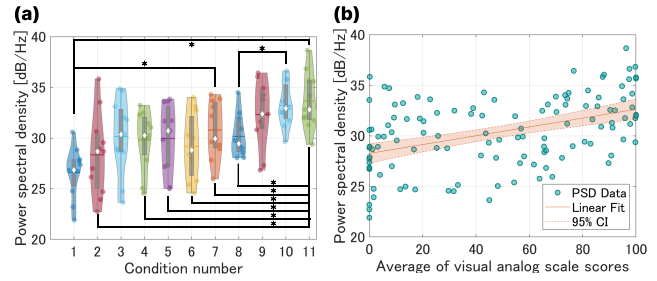


Figure 3: Results of PSD Analysis. (a) PSD of each stimulus conditions. * indicates statistically significant ($p < 0.05$). (b) Correlation between reported flickering magnitude and PSD ($r = 0.443$, $p < 0.001$). Abbreviations: PSD–power spectral density.

4 DISCUSSION

The results of the present study showed that the PSD of pupillary oscillations increased in correspondence with the differences of equiluminant color flickering. This result indicates that the perception of flickering color change can be estimated from the pupillary response. A limitation of this study is that we used combinations of colors in the confusion line of deutan color vision; thus, whether our method is applicable to other visions, such as protan vision, requires further discussion.

5 CONCLUSION

Pupillary oscillation reflected subjective color perception by flickering equiluminant pseudo-isochromatic color patches. In the future, it is necessary to start development as a color vision measurement device based on the method of color vision estimation from the trend by pupillary response. This device can estimate subjective color distinguishing ability by pupillary response, and pursue a more feasible color testing method.

Some VR devices have eye trackers installed out of the box (e.g. HTC VIVE Pro Eye, PlayStation VR2). With the widespread use of these devices, our method may gain wide application in the estimation of the color perception of users.

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